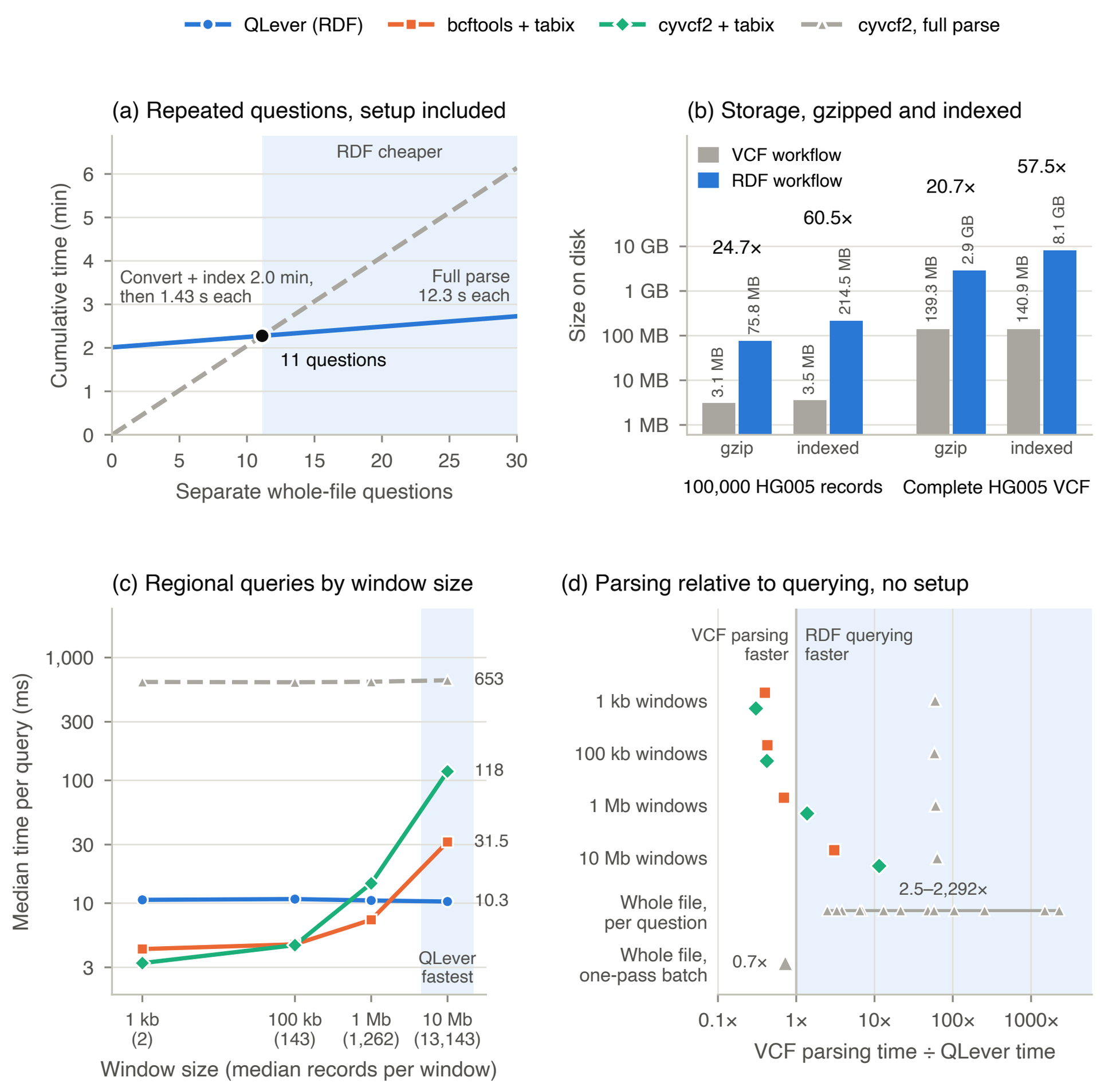
QLever (RDF)
bcftools + tabix
cyvcf2 + tabix
cyvcf2, full parse
(a) Repeated questions, setup included
RDF cheaper
Convert + index 2.0 min,
then 1.43 s each
Full parse
12.3 s each
11 questions
Cumulative time (min)
Separate whole-file questions
(b) Storage, gzipped and indexed
VCF workflow
RDF workflow
Size on disk
24.7×
60.5×
20.7×
57.5×
3.1 MB
75.8 MB
3.5 MB
214.5 MB
139.3 MB
2.9 GB
140.9 MB
8.1 GB
gzip
indexed
gzip
indexed
100,000 HG005 records
Complete HG005 VCF
(c) Regional queries by window size
Median time per query (ms)
653
118
31.5
10.3
QLever
fastest
1 kb
(2)
100 kb
(143)
1 Mb
(1,262)
10 Mb
(13,143)
Window size (median records per window)
(d) Parsing relative to querying, no setup
VCF parsing
faster
RDF querying
faster
1 kb windows
100 kb windows
1 Mb windows
10 Mb windows
Whole file,
per question
Whole file,
one-pass batch
2.5–2,292×
0.7×
VCF parsing time ÷ QLever time